Jose Andres Daza Gallego

Computer Engineering Student · Technical Diploma in Programming

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Summary

Computer Engineering student with hands-on experience in cross-platform software development, server administration, and laboratory management. Creator of **SGI LAB MANAGER**, a full-featured lab management system currently in production, built with Flutter and Django, and of **Vigía**, a satellite-based wildfire early-warning system. Proven ability to lead projects end-to-end, collaborate with multidisciplinary teams, and deliver technical solutions in real-world environments.

Technical Skills

Languages	Python, Java, JavaScript, TypeScript, Dart, C++, C#
Front-End	HTML, CSS, Flutter, D3.js
Back-End	Django, Celery, Node.js, PHP, pytest
Databases	MySQL, PostgreSQL, PostGIS, MongoDB, Redis
DevOps / Infra	Git, Docker, Kubernetes, Nginx, Gunicorn, Cloudflare
3D / VR	Unity, Blender, OpenXR, Meta XR SDK
AI / Data	scikit-learn, Groq (Llama 3.1), Google Cloud TTS
Methodologies	Scrum
Operating Systems	Linux (Ubuntu, Kali, Arch), Bash, Windows

Soft Skills: Leadership, effective communication, problem-solving, teamwork.

Experience

SGI LAB MANAGER
February 2024 – Present

Creator & Lead Developer

- Designed, developed, and launched the first version of a full lab management system for Politécnico Colombiano Jaime Isaza Cadavid.
- Cross-platform architecture with **Flutter/Dart** (Android, Web, Windows, Linux) and **Django REST Framework** backend on PostgreSQL 15.
- Full DevOps ownership: Docker Compose, Nginx, Gunicorn, Cloudflare Zero Trust Tunnel, automated encrypted backups (GPG AES-256), and Telegram monitoring.
- Modules: GHS chemical safety inventory with AI-powered SDS extraction (Groq/Llama 3.1), loans, weekly scheduling, 6 PDF report types, public QR forms, and 3D printing orders (multi-part orders, partial payments, QR tracking).
- 450+ Django tests, 60+ pytest e2e, and 190+ Flutter tests. v1.2.0 (Nov 2026). End-to-end ownership: planning, architecture, development, deployment, and ongoing maintenance.

Politécnico Colombiano Jaime Isaza Cadavid
February 2024 – Present

Laboratory Monitor

- Part of the two-person lab monitor team; previously, support was handled exclusively by the two full-time lab technicians.
- Provide daily technical support to students and faculty, managing preventive and corrective maintenance of equipment and specialized software.
- Operational needs identified in this role led to the development of **SGI LAB MANAGER**, now

in active production.

Politécnico Colombiano Jaime Isaza Cadavid *January 2024*
Server Administrator *Present*

- Configured and administer the institutional server hosting the production environment of **SGI LAB MANAGER**: Docker Compose, Nginx, Gunicorn, and Cloudflare Zero Trust Tunnel.
- Maintain a personal homelab that mirrors the production infrastructure, used as a development and testing environment before deploying changes to the institutional server.

Immersive Learning Environments Research Group *2023*
VR Researcher & Developer — Politécnico Colombiano Jaime Isaza Cadavid *August 2026*

Project concluded due to loss of institutional research support.

- Co-founding member of the research group since its formation in October 2023.
- Developed immersive 3D environments in Unity and VRChat with Meta Quest 2, including the navigable Bloque R27 model and the Politécnico virtual campus.
- The 3D assets produced during this project serve as the foundation for the ongoing Virtual VR Laboratory.

Notable Projects

SGI LAB MANAGER

FoxyYTs/SGI_LAB_MANAGER_FRONTEND

Cross-platform lab management platform (Android, Web, Windows, Linux) built with Flutter/Dart and Django REST Framework. Own infrastructure: Docker Compose, Nginx, Gunicorn, PostgreSQL 15, Cloudflare Zero Trust Tunnel, and automated encrypted backups (GPG AES-256) with Telegram monitoring. Modules: GHS inventory with AI-powered SDS extraction (Groq/Llama 3.1), loans, weekly scheduling, 6 PDF report types, and public QR forms, and 3D printing orders. 450+ Django tests · 60+ pytest e2e · 190+ Flutter tests. v1.2.0 (Nov 2026).

Vigía — Wildfire Alerts

FoxyYTs/Vigia Production site

Wildfire early-warning system for Colombia, built solo as my capstone project. Ingests NASA FIRMS and INPE hotspots every 3 hours (over 880 thousand records) and cross-references UNGRD and IDEAM historical data; alerts are grouped and delivered through Telegram. Recurrent-zone model with scikit-learn across 1,122 municipalities. Django REST, PostGIS and Celery, with a Flutter web and Android app. In production with Docker and Cloudflare Tunnel, CI on GitHub Actions and 240+ tests.

Interactive Academic Curriculum — PCJIC Rionegro

FoxyYTs/pensum-interactivo Live site

Interactive curriculum visualizer for all 8 degree programs at the Rionegro campus of Politécnico Colombiano Jaime Isaza Cadavid. Directed graph with D3.js showing courses, prerequisites, credits, and completion percentage, with `localStorage` persistence. PWA deployed on GitHub Pages with automated CI/CD via GitHub Actions. Adapted fork of a Chilean open-source project, with full terminology migration and new features.

Virtual VR Laboratory

In development — June 2026

Successor to the Immersive Learning Environments research group. Virtual lab platform for the Politécnico's Meta Quest 2 headsets, reusing the Bloque R27 spaces digitized during the research group as the setting for interactive lab sessions. 4 modules (Physics, Waves & Optics, Chemistry, Electricity & Magnetism) with 12 guided practices plus one VR-exclusive practice per module. Built with Unity 6.3 LTS, C#, OpenXR, and Meta XR SDK. Three modes: Meta Quest 2 standalone, PC VR, and desktop.

Education

B.Eng. in Computer Engineering — in progress, 9th semester *August 2022 — Expected 2027*
Politécnico Colombiano Jaime Isaza Cadavid

Technical High School Diploma in Programming *February 2015 — December 2020*
I.E. INEM José Félix de Restrepo

Additional Education & Certifications

- **Git & GitHub** — Platzi (Dec 2025)
- **Software Engineering Fundamentals** — Platzi (Dec 2025)
- **MongoDB for SQL Professionals** — MongoDB Academy (Nov 2025)
- **Linux Essentials** — LPI / Cisco (Nov 2025)
- **Django** — Platzi (Aug 2025)
- **Python** — Platzi (Aug 2025)
- **Certified Kubernetes Administrator (CKA) with Practice Tests** — Udemy / KodeKloud (Apr 2025)
- **PL DEVOPS** — SoftServe Academy (Jun 2025)
- **Java SE Object-Oriented Programming** — Platzi (Sep 2023)
- **Object-Oriented Programming** — Platzi (Apr 2023)
- **Personal Development G8 - ONE** — Alura & Oracle (Jan 2025)
- **Game Development** — Platzi (Jan 2021)

Languages

Spanish — Native **English** — Intermediate (reading and technical proficiency)

References

Luz Delly Jaramillo Carmona — *Lab Supervisor, Politécnico Colombiano JIC*
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Ángela Viviana Martínez Betancur — *Lab Supervisor, Politécnico Colombiano JIC*
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Ariel Yesid Aristizábal Álzate — *Lab Supervisor, Politécnico Colombiano JIC*
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Phone numbers available upon request.

Interests & Hobbies

Electronics & IoT · **Motorcycle Mechanics** · **Game Development** · **DIY Projects** · **Cybersecurity**